



Nuclei™ N200 Series Processor Core Check List

Nuclei

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Contents

1	Copyright Notice	1
2	Contact Information	2
3	N200 Series Quick-Start	5
4	Appendix	7

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Contact Information

Should you have any problems with the information contained herein or any suggestions, please contact Nuclei System Technology by email support@nucleisys.com, or visit “Nuclei User Center” website <http://user.nucleisys.com> for supports or online discussion.

List of Figures

List of Tables

N200 Series Quick-Start

N200 series product has the complete set of documents, which are listed at below table, to help user quickly familiarize it. And this release is for Release-2.5.0 of N200 series product.

Name	Version	Descriptions
<Nuclei_RISC-V_ISA_Spec.pdf>	v2.6.2	These documents systematically introduce the Nuclei RISC-V architecture specification, including: • Instruction set, • Privileged architecture, • CSR registers, • Low-power architecture scheme, • And the interrupts, etc.
<Nuclei_N200_Series_Databook.pdf>	v1.6.0	This doc as Databook introduces the IP's: • Features, • Components, • Interfaces, • The configuration options, • Other overall introductions, etc.
<Nuclei_Processor_Integration_Guide.pdf>	v1.5.0	This doc introduces: • How to configure the IP options and generate the Verilog RTL codes and view it. • How to run the basic CCT (Customer Confidential Tests) with the attached basic test-bench environment. • How to integrate the CPU core into the customer SoC. • How to generate FPGA bitstream for SoC prototype (including RISC-V Core) . • How to run the synthesis. • Other overall introductions, etc.
<Nuclei_NICE_Extension.pdf>	v1.2.0	This doc introduces how to use NICE interfaces of Nuclei core to implement Customer Instructions.

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Table 3.1 – continued from previous page

Name	Version	Descriptions
<Nuclei_TEE_Architecture.pdf>	v1.0.1	This doc introduces TEE features of Nuclei core and how to use this feature.
<Nuclei_Eval_SoC_Intro.pdf>	v1.1.0	This document introduce the prototype SoC for evaluation purpose, this prototype SoC included bus, common peripherals, etc.
<Nuclei_RISC-V_ISA_Spec-Chinese.pdf>	v2.1.1	This document is Chinese version of <Nuclei_RISC-V_ISA_Spec.pdf>, while it only covers the N Series Core (RV32) and the content may be out of date. It is convenient for users who want to read Chinese document.

Nuclei also offers online download of IDE/Toolchain, on line documents and open source SDK for users, please find the portals in below table:

Name	Portal	Descriptions
NucleiStudio & Toolchain	https://www.nucleisys.com/download.php	It gets the NucleiStudio IDE and Toolchain download link.
Nuclei SDK	https://github.com/Nuclei-Software/NMSIS	It is the open source project on Github. Also it gets the Gitee repository: https://gitee.com/Nuclei-Software/NMSIS Documents: https://doc.nucleisys.com/nuclei_sdk/
Nuclei Linux SDK	https://github.com/Nuclei-Software/nuclei-linux-sdk	It is the open source project on Github. Also it gets the Gitee repository: https://gitee.com/Nuclei-Software/nuclei-linux-sdk
NMSIS	https://github.com/Nuclei-Software/NMSIS	It is the open source project on Github. Also it gets the Gitee repository: https://gitee.com/Nuclei-Software/NMSIS Document: https://doc.nucleisys.com/nmsis/
RISC-V MCU Website	https://www.rvmcu.com/	The webpage as the embedded RISC-V MCU ecosystem website, And it is Chinese content.
Nuclei Board Labs	https://doc.nucleisys.com/nuclei_board_labs/	The webpage as the embedded software labs ecosystem website, for Nuclei MCU development board. Currently it is English content.

- **Nuclei RISC-V IP Products:** <https://www.nucleisys.com/product.php>
- **Nuclei Spec Documentation:** <https://nucleisys.com/download.php#spec>
- **Nuclei RISC-V Tools and Documents:** <https://nucleisys.com/download.php>
- **Nuclei Prebuilt Toolchain and IDE:** <https://nucleisys.com/download.php#tools>
- **NMSIS:** <https://github.com/Nuclei-Software/NMSIS>
- **Nuclei SDK:** <https://github.com/Nuclei-Software/nuclei-sdk>
- **Nuclei Linux SDK:** <https://github.com/Nuclei-Software/nuclei-linux-sdk>
- **Nuclei Software Organization in Github:** <https://github.com/Nuclei-Software/>
- **RISC-V MCU Organization in Github:** <https://github.com/riscv-mcu/>
- **RISC-V MCU Community Website:** <https://www.rvmcu.com/>
- **Nuclei riscv-openocd:** <https://github.com/riscv-mcu/riscv-openocd>
- **Nuclei riscv-gnu-toolchain:** <https://github.com/riscv-mcu/riscv-gnu-toolchain>